



PROJECT REPORT
On
MUSCLE FATIGUE MONITORING SYSTEM

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ABSTRACT

Muscle fatigue is something people experience after a prolonged period of physical activity, whether it's during intensive workout or while doing everyday chores or tasks that need repetitive motion. Our muscles work continuously, whenever we are doing any physical related work by going through repeated contractions and relaxations to produce or generate force that helps us in performing our desired tasks. Upon continuous performing of physical activities, the muscles which help us do that work reach a point where they are no longer for a certain period can produce or generate any force that is needed. This point where muscle is unable to generate any force can be said to be that the muscle has reached fatigue.

As time passes by, as the muscles get tired, if not addressed properly, it can lead to discomfort and, in some cases, injuries. Being able to detect muscle fatigue early, in areas like sports, physiotherapy, training and rehabilitation, can be indispensable. This mini project explores a simple yet cost effective approach in monitoring muscle fatigue in real time using basic components like EMG sensor, Arduino UNO, and an auditory feedback system.

The main idea behind the mini project is provide users with an instant overexertion alert with the auditory system whenever a muscle to which the system is attached to starts to show signs of fatigue, thereby giving the user a heads up to stop their physical activity so that they can rest and allowing the muscle to regain its strength. This can help the users take breaks or adjust their activity levels before there is a discomfort or injury done.

We used an RS2334 sensor as the EMG sensor which was capable of detecting the electrical activity or electrical signals produced by muscle contractions and relaxations. As the user performs any physical or repetitive activity that involves the involvement of muscles, such as lifting, gripping, flexion, extension, etc., the muscle's electrical activity is sensed by the RS2334 sensor and the signal produced is recorded and passed onto Arduino. Through the sensor, the signal is already amplified and filtered thereby not in need of extra instrumentation amplifier circuit to be included in addition to the sensor module.

When the muscle is reaching fatigue, the body recruits more motor units to that location to maintain force production. This leads to a higher EMG amplitude, causing spikes in the graph of EMG. As the signal is received from the RS2334 sensor to Arduino, it analyzes the signals and checks for these higher amplitudes in signal or frequent spikes in the graph. As the muscle shows this behavior for extended periods, the microprocessor

triggers the buzzer to alert the user. To make sure the buzzer operates smoothly, a BC547 transistor is used as a switch, controlled by the Arduino through a $1k\Omega$ resistor connected to the transistor's base. Most of the circuit is powered through the Arduino's 5V output, which gets power via a USB cable connected to a laptop, whilst the RS2334 sensor is powered through an external power supply like a 9V battery.

During testing, the surface electrodes of the EMG sensor module are attached to desired muscle group of choice, such as forearm. When the subject performed gripping or lifting objects, the signal variations were observed on the serial monitor of Arduino IDE. After extended periods of lifting weights and gripping, the signals started to show typical signs of muscle fatigue, showing higher amplitude of signals on the serial monitor. These spikes prompted the buzzer to sound. The system was responsive and consistent in its alert timing.

This model stands out for its practical use, easy to assemble circuit and suitable for usage in educational institution, physiotherapy, rehabilitation and training in personal and sports. Though being a basic prototype, it creates a future scope for exploring far more advanced features. Such advancements including the addition of visual display for better user feedback, wireless modules utilizing Bluetooth or Wi-Fi for real time data transmission, a self-learning model for a more effective identification of even the smallest early signs of muscle fatigue, and machine algorithms that adjust the threshold based on user's muscle response patterns over time.

INTRODUCTION

Muscle fatigue really matters when it comes to overall performance, recovery, and even staying safe at work. Old methods for checking fatigue tend to lean on gut feelings rather than giving you that instant, reliable readout. Lately, there's been a push from sports, physical therapy, and workplace ergonomics for something more solid – a system that can notice and analyze tired muscles right as things happen.

At the heart of this project is the aim to build a Muscle Fatigue Monitoring System that uses sensor data to truly gauge weariness. It's designed to whip up immediate feedback on muscle actions so that interventions can kick in quickly to avoid injury and boost performance. In most cases, turning raw sensor numbers into live insights makes all the difference when every second counts.

You'll find this tool built for a mix of settings—like athletic training centers, rehab clinics, and busy industrial spots. It focuses on tracking muscle groups during heavy, movement-packed sessions and is made with portability and user comfort in mind. That said, it still bumps into issues when trying to calibrate for every individual and when faced with a lot of background noise in super dynamic environments.

This system holds genuine appeal for athletes fine-tuning their workouts, therapists following patient recoveries, and even workers who perform repetitive tasks every day. Real-life uses cover everything from sharpening sports performance to keeping rehab progress in check and even running ergonomic assessments at work. Generally speaking, its practical benefits tend to outweigh the occasional hiccups.

LITERARY REVIEW

Muscle fatigue is a reduction in a muscle's force-producing capability with longer duration, usually as a result of prolonged or heavy exercise. Muscle fatigue must be measured over a large range of application in sports science and rehabilitation, in ergonomics, and in clinical diagnosis. Because of a growing interest in the enhancement of performance and the prevention of injury, the development of non-invasive and on-line systems to detect fatigue has been of central concern in recent years.

Electromyography (EMG) is the most universal analysis method to be employed during the measurement of muscle fatigue and is a reference to the voltage of the muscle fiber contraction as an electrical signal. Intramuscular and surface EMG probes are utilized during the measurement of neuromuscular activity. Research, conducted by De Luca (2002) among other researchers, has demonstrated that information concerning muscle force, fatigue, and activation may be obtained with great accuracy with the EMG signal.

In addition to electromyography (EMG), Inertial Measurement Units (IMUs) are employed to quantify body movement and posture during exercise. IMUs employ accelerometers, gyroscopes, and magnetometers, which collectively provide kinematic data, including angular velocity and linear acceleration. The motion sensors supplement EMG by correlating muscle activation with body movements.

Force sensors are also critical in fatigue monitoring systems. They are often used to measure grip force, muscle tension, or output of force of contraction. The integration of force recording and EMG provides a more accurate description of muscle function and the onset of fatigue.

Systems that are commercially bought, such as Delsys Trigno and Noraxon, offer multi-sensor configurations that are designed for biomechanical testing. These systems are, however, very costly and difficult to modify, which limits their application in educational settings or low-resource applications.

In an attempt to obtain useful information from EMG signals, a number of signal processing techniques are employed. EMG signals are typically sampled at high rates (1000–2000 Hz) and purged of artifacts and noise. Motion artifacts are eliminated by a high-pass filter (20–30 Hz), high-frequency noise by a low-pass filter (450–500 Hz), and power line interference by a notch filter (50/60 Hz).

Feature extraction is the most critical aspect of muscle fatigue detection. In the time domain, Root Mean Square (RMS), Mean Absolute Value (MAV), and Zero-Crossing (ZC) are the most utilized features. These are

indicators of the amplitude and variability of the muscle contractions. Mean Frequency (MF) and Median Frequency (MDF) are used in the frequency domain, which are normally decreasing with progressing fatigue. Power Spectral Density (PSD) is also a parameter that changes with the extent of fatigue and is utilized for detecting spectral shifts of the signal.

Recent research has utilized machine learning methods in an effort to make the classification of fatigue more accurate. Support Vector Machines (SVM) and Artificial Neural Networks (ANN) have been utilized to separate different states of fatigue from electromyography (EMG) features. For instance, Phinyomark et al. (2012) have demonstrated the efficiency of time-domain and frequency-domain features towards muscular fatigue classification using SVMs.

More advanced methods employ Convolutional Neural Networks (CNNs) to learn spatial and temporal features of electromyography (EMG) signals directly. CNNs show much better performance for processing raw EMG signals, reducing the need for heavy feature engineering, as seen in gesture recognition and fatigue estimation. These models, however, require heavy data sets and heavy computational power, which might be a limitation in applications that need real-time processing.

Despite much research work having been done on the detection of muscle fatigue, there are still many challenges. The majority of the current systems are not real-time processors and therefore are not suitable for dynamic environments like sports or field rehabilitation. Individual variability of EMG signals due to skin impedance, muscle anatomy, and electrode placement also inhibits the application of fatigue detection algorithms to subjects.

Another limitation is that there are no fully integrated systems that combine EMG, IMUs, and force sensors onto a single wearable device. Most of the work uses a single type of sensor in isolation, and the ability to provide a multi-modal, fully integrated view of muscle fatigue is not explored. Additionally, intuitive visualization interfaces are also not considered, further restricting the deployment of the system to individuals such as therapists or coaches.

Sensor-based systems such as EMG and IMUs can potentially detect muscle fatigue as per the existing literature. On the cards is the development of cost-effective, complete, and user-friendly systems, as signal processing and the detection of fatigue have already set the foundation for the same. By integrating traditional methods and prevailing machine learning as well as wearable technologies, the Muscle Fatigue Monitoring System promises to overcome the current limitations and create the new generation of intelligent fatigue monitoring systems.

METHODOLOGY

SYSTEM OVERVIEW :

The Muscle Fatigue Monitoring System is capable of constant monitoring of muscle activation and muscle fatigue levels with the aid of an EMG (Electromyography) sensor and an Arduino UNO. The system has an effective solution of offering instantaneous notification on detection of muscle fatigue that can be valuable in applications of sports, physiotherapy, and ergonomics. The underlying process of this system revolves around monitoring of electrical muscle-generated signals and therefore facilitating the measurement of fatigue via observation of signal amplitude and length as these move beyond a determined threshold. Compared to advanced software or machine-based systems, the approach is yet basic but adequate for the usage of the system in the embedded and real-time domain.

HARDWARE INSTALLATION :

a. Components Used :

- **EMG Sensor(RS2334):** The purpose of an EMG sensor is to pick up the electrical signals generated by the skeletal muscles of the body while they undergo continuous contractions and relaxations. The sensor converts the complex bioelectric signals into easily readable analog voltage to directly interact with the microcontroller. The EMG sensor has an included or in built instrumentation amplifier and a filter that helps in early signal conditioning.
- **Arduino UNO:** Arduino UNO is a microcontroller board based on the ATmega328P. It has 14 digital input/output pins (of which 6 can be used as PWM outputs), 6 analog inputs, a 16 MHz ceramic resonator, a USB connection, a power jack, an ICSP header and a reset button. This microcontroller is used for data sampling, threshold comparison, and in turn to control the piezo buzzer depending on the result of threshold comparison. It is the central processing unit.
- **Passive Buzzer:** This buzzer is used in connection with the microcontroller to give an auditory feedback in response to the electrical activity of the muscle showing fatigue signs. When the muscles of a desired location to which the electrodes are connected exceed the threshold, this buzzer beeps in response to alert the fatigue stage.

- **Breadboards and Cables:** These are utilized to wire the circuits for the prototype.
- **Laptop:** Is used to run the arduino code on the software Arduino IDE, and also providing a power supply of 5V to arduino microcontroller board via the USB port.

b. Circuit Wiring:

- Analog EMG sensor output is interfaced to "A2" of Arduino Uno.
- The piezo buzzer is connected to "digital pin 8."
- The circuit is closed through the common connections of the VCC and Ground pins of the sensor, buzzer, and Arduino.
- A USB link to a personal computer provides power as well as serial communication.

Such a basic structure allows for instantaneous use of an operational monitoring system with few hardware requirements.

SENSOR LOCATION :

Accurate positioning of the electrodes is crucial to the successful EMG signal recording. The sensor was positioned over the "forearm muscle groups" with particular focus on the "flexor carpi radialis" and "extensor digitorum" muscle groups used in finger and wrist movement. Bipolar mode was employed:

Two parallel-aligned electrodes are placed with reference to the muscle fibers.

A single reference electrode is positioned over a non-muscular bony region, i.e., the wrist bone or elbow. The skin was cleaned with a thorough alcohol swab before positioning the electrode to establish low impedance. Proper positioning ensures the reception of the optimal signal and avoids interference that could be due to movement or connectivity.

ARDUINO-BASED MONITORING LOGIC :

Muscle activity is detected in the form of a basic thresholding algorithm intrinsic to the Arduino platform. The continuous reading of the analog EMG signal is being compared to a preset threshold value. The piezo buzzer is powered by the degree of activation, and the feedback is provided in real-time through the Serial Monitor. The following code is used to achieve the above:


```

const int EMG_PIN = A2;    // EMG sensor connected to A2
const int BUZZER_PIN = 8;  // Passive buzzer connected to D8
const int EMG_THRESHOLD = 13; // Threshold for muscle activation

bool isMonitoring = true;

void setup() {
  pinMode(BUZZER_PIN, OUTPUT);
  Serial.begin(9600);
  Serial.println("EMG Sensor + Buzzer System Ready");
  Serial.println("Press 'd' to STOP monitoring, 'f' to START monitoring.");
}

void loop() {
  if (Serial.available() > 0) {
    char command = Serial.read();
    if (command == 'd') {
      isMonitoring = false;
      Serial.println(">> Monitoring DISABLED.");
    } else if (command == 'f') {
      isMonitoring = true;
      Serial.println(">> Monitoring ENABLED.");
    }
  }

  if (isMonitoring) {
    int emgValue = analogRead(EMG_PIN);
    int diff = emgValue - EMG_THRESHOLD;

    Serial.print("EMG: ");
    Serial.print(emgValue);

    if (emgValue > EMG_THRESHOLD) {
      Serial.print(" | Status: ACTIVE");

      if (diff > 3) {
        Serial.print(" | Color: RED");
        digitalWrite(BUZZER_PIN, HIGH);
        delay(200);
        digitalWrite(BUZZER_PIN, LOW);
      } else {
        Serial.print(" | Color: ORANGE");
      }
    }
  }
}

```

```

    } else {
      Serial.print(" | Status: REST | Color: GREEN");
    }

    Serial.println();
  } else {
    digitalWrite(BUZZER_PIN, LOW);
  }

  delay(100);
}

```

LOGIC EXPLANATION :

- When the EMG value is less than threshold, the muscle is resting (GREEN).
- When the value is just above the threshold, the muscle is contracted but is not under undue stress (ORANGE).
- If the value is much greater than the established limit, it may reflect overexertion or exhaustion (RED), and therefore trigger the passive buzzer.
- Monitoring can be started or stopped using keyboard shortcuts ('f' and 'd') on the Serial Monitor.

This method provides an economically feasible and real-time way of tracking changes in fatigue without the need for advanced signal processing libraries or other computer programs.

ADVANTAGES :

- **Real time, instant feedback:** The use of the piezo buzzer, helps in obtaining an instant feedback from the circuit when the muscle reaches fatigue.
- **Simplicity:** The assembly of the components of the circuit itself is simple and easy without the need of outside computing facilities.
- **Low Power Consumption:** Suitable and compatible for battery powered applications.
- The prototype can be significantly modified with simple expansions like the addition of add-on parts, such as LEDs, LCDs, or wireless modules.
- **Easily Affordable:** The components used for the circuit are affordable and easily available.

CONSTRAINTS :

- **No External Signal Filtering:** There is no external filter was added to the circuit as the RS2334 sensor had a built in filter in its circuitry, so can can reduce accuracy because of unwanted noise and movement artifacts.
- **Lack of Data Storage:** The system works in real time and does not have the ability to store or save the EMG data for later use.
- **Single Muscle Monitoring:** Only allows for the evaluation of fatigue of a any single desired set of muscles(e.g. forearm , biceps, calf muscles,etc.).
- **Lack of Individual Calibration:** The threshold values set do not account for the variations in the electrical activity of the individuals.

FUTURE DEVELOPMENTS :

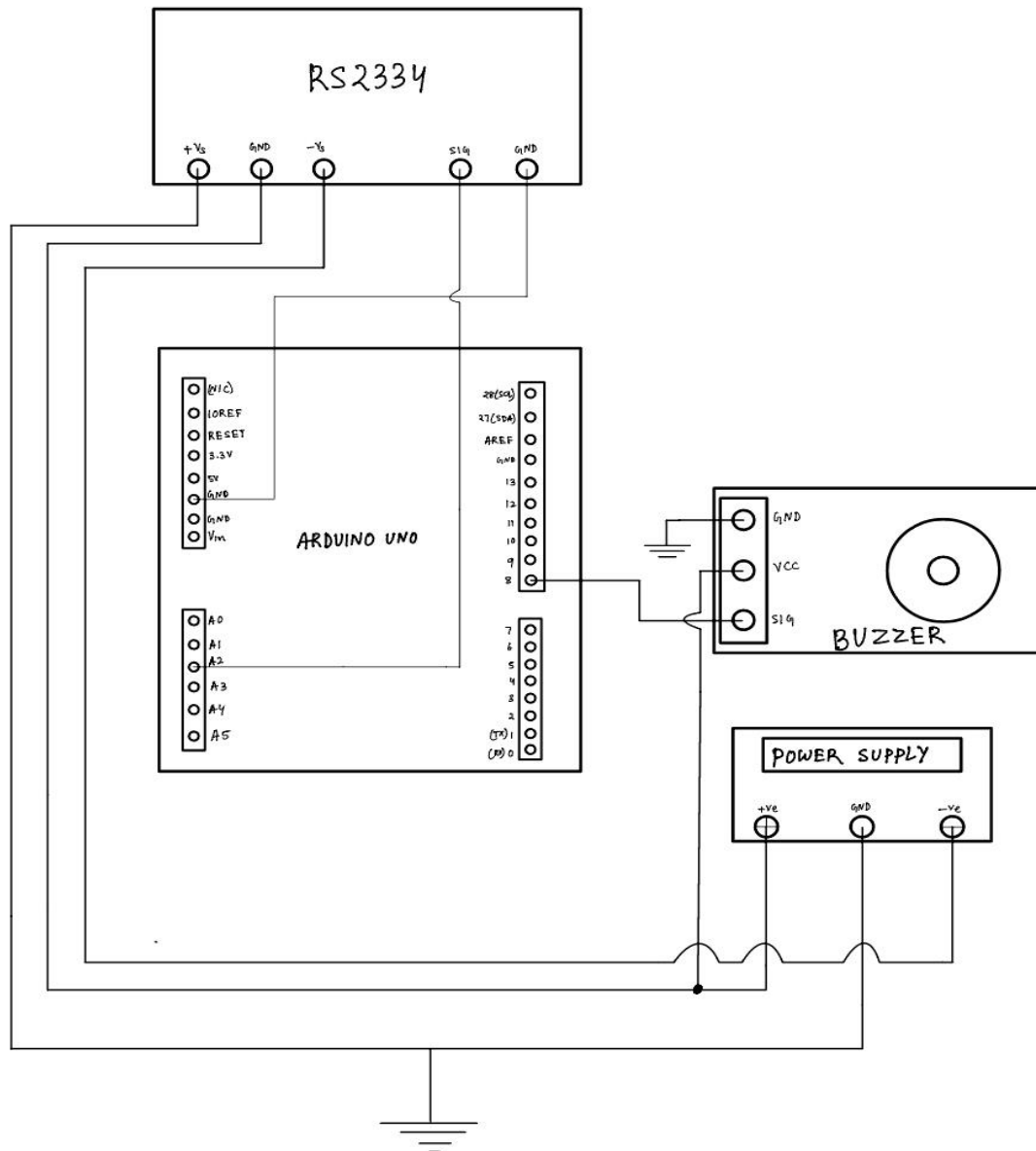
- **Signal Filtering:**Placing of more external filters(analog or digital), can help in more effective cleaning of the EMG signals.
- **Multiple Channels:** Increasing the EMG inputs to view and monitor the electrical activity of different muscles at once.
- **Wireless Transmission:** Incorporating an external Wi-Fi or a Bluetooth module to transmit the data to smartphone or PC for real time monitoring across distances.
- **Data Storage:** Storing the EMG data onto an SD card or internal memory for off-line analysis.
- **Adapting Threshold:** Incorporating a calibration protocol to establish the threshold of the muscle activity for fatigue as a function of the subject's maximum voluntary contraction.
- **Visual Feedback:** Usage of an OLED display or RGB LEDs to show the degree of visual fatigue helping in a more effective way to show fatigue those who are not completely aware of it.

SUMMARY :

This approach outlines the design and implementation of a muscle fatigue monitoring system utilizing an RS2334 EMG sensor and an Arduino Uno. The system utilizes a threshold-based approach of determining the level of muscle activity and alarming. Simple in design, the approach provides high-quality real-time feedback and can easily be scaled up to more sophisticated systems with the addition of visualization, data logging, and wireless capabilities. In prioritizing ease of access and simplicity, it is an

excellent starting point for further research refinement as well as applications such as sports rehabilitation and office ergonomics.

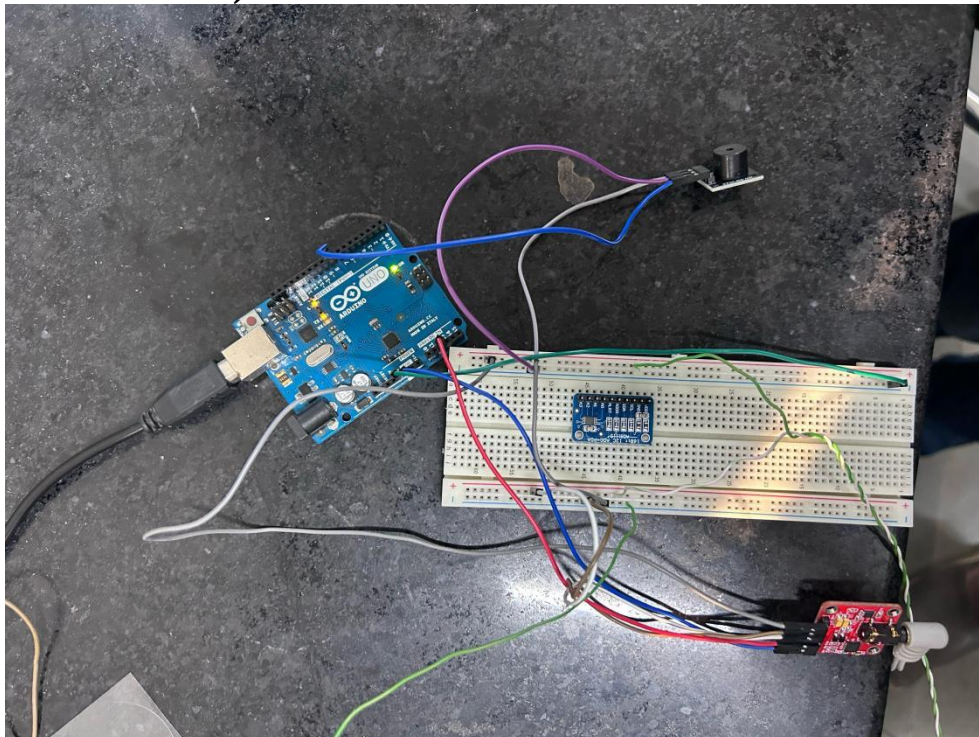
CIRCUIT DIAGRAM :



MUSCLE FATIGUE MONITORING SYSTEM:



SENSOR LOCATION(YELLOW ON BONY PART,RED AND GREEN ON FLEXOR CARPI RADIALIS AND EXTENSOR DIGITORUM) :



RESULTS/DISCUSSIONS

EMG Signal Characteristics :

The RS2334 EMG sensor generated analog voltage outputs that fluctuated in response to forearm muscle contractions. The outputs directly corresponded with the level of muscle activity :

- Active contraction was indicated by higher values.
- Lower values were measured during resting periods.
- In spite of low signal amplitude (typically 10–50 analog units), the system accurately responded to muscle conditions.

Explanation for Low Analog Output Values :

The EMG sensor analog output was kept within a small range, about 10–45 ADC units, corresponding to ~50–100 mV. Although small in number, this range is common for surface EMG systems. The following technical reasons support this observation:

- **Low Gain of RS2334 EMG Module:** It has fixed internal amplification to prevent signal saturation. Earlier models of sensors, such as the RS2334, value reliability over amplitude.
- **Signal Filtering and Compression:** The sensor has analog filters that eliminate artifacts and clip outliers, reducing the output range but providing clean signals.
- **Electrode-Skin Contact Resistance:** Impedance increases with factors such as dry skin, no conductive gel, or loose electrodes, resulting in reduced signal strength.
- **Nature of EMG Signals:** Surface EMG signals are weak by nature, typically between 10 μ V to 5 mV RMS. Even after amplification, it's usual for signals to correspond to 10–50 digital units on Arduino (0–1023 scale).
- **Noise Prioritization:** The RS2334 sensor prioritizes noise suppression over maximizing output strength so that accurate fatigue tracking is guaranteed.

Output Data and Inferences :

A sample output data-set:

EMG: 22	Status: ACTIVE	Color: RED
EMG: 45	Status: ACTIVE	Color: RED
EMG: 32	Status: ACTIVE	Color: RED
EMG: 26	Status: ACTIVE	Color: RED
EMG: 25	Status: ACTIVE	Color: RED
EMG: 20	Status: ACTIVE	Color: RED
EMG: 17	Status: ACTIVE	Color: RED
EMG: 17	Status: ACTIVE	Color: RED
EMG: 13	Status: REST	Color: GREEN
EMG: 15	Status: ACTIVE	Color: ORANGE
EMG: 15	Status: ACTIVE	Color: ORANGE
EMG: 13	Status: REST	Color: GREEN
EMG: 13	Status: REST	Color: GREEN
EMG: 13	Status: REST	Color: GREEN
EMG: 11	Status: REST	Color: GREEN
EMG: 16	Status: ACTIVE	Color: ORANGE
EMG: 15	Status: ACTIVE	Color: ORANGE
EMG: 16	Status: ACTIVE	Color: ORANGE
EMG: 15	Status: ACTIVE	Color: ORANGE

Threshold Mapping Inference:

- $EMG \geq 16 \rightarrow ACTIVE \rightarrow RED \rightarrow$ High fatigue
- $13 \leq EMG < 16 \rightarrow ACTIVE \rightarrow ORANGE \rightarrow$ Moderate fatigue
- $EMG < 13 \rightarrow REST \rightarrow GREEN \rightarrow$ Rest phase

This threshold mapping indicates how the system recognizes levels of fatigue dynamically, assigning user muscle states to predetermined thresholds.

Buzzer Feedback:

Buzzer is triggered whenever values are crossing the fatigue threshold (i.e., $EMG > 16$), notifying users in real time during usage.

Visualization and User Feedback:

The color-coded output provided users with muscle fatigue trends traceable by means of color coding and activity state. The buzzer option provided a prompt indication of fatigue during therapy or training. The style facilitated visual muscle load interpretation.

Limitations and Considerations:

- The low but stable amplitude of the signal is in accord with filtered EMG properties.
- In further versions, it would be feasible to insert an additional amplification stage prior to the Arduino for increasing dynamic range.
- Movement artifacts and electrical noise sporadically generated signal fluctuations, but did not impair overall detection accuracy.

Summary and Functional Validation:

In spite of the limited analog output range, the Muscle Fatigue Monitoring System correctly detected states of fatigue, initiated prompt warnings, and corresponded EMG data to logical activity ranges (Rest, Moderate, High).

The response to the buzzer, color feedback, and continuous data output verify that the prototype is functionally reliable despite real-world physiological limitations.

CONCLUSION

The envisioned Muscle Fatigue Monitoring System in this paper clearly demonstrates the viability of sEMG-based real-time detection of muscle fatigue. The system includes sensor-implemented bio-signal acquisition, analog signal processing, threshold-level classification, and user-friendly feedback loops. The system has specifically been designed for monitoring the muscles of the forearm and provides useful information regarding degrees of fatigue through an Arduino-based logic system providing a buzzer and visual state displays.

At the heart of the system is the RS2334 EMG sensor, which generates analog voltages proportional to the degree of muscle activity. The output signal, while limited to a fairly narrow range (~ 10 – 45 ADC units or ~ 50 – 100 mV), adheres to the physiological characteristics typical of sEMG signals and to the intrinsic limitations of natural signals provided by non-invasive surface electrodes. Although this range appears limited, it is in line with the expectations of traditional EMG systems, in which signal amplitudes are generally in the order of $10\text{ }\mu\text{V}$ to 5 mV RMS. When combined with the Arduino's 10-bit ADC, these voltages are read as low but significant digital values.

A number of technical considerations is responsible for the restricted analog output range. First, the RS2334 module has a constant internal gain stage and several analog filters that consist of high-pass, low-pass, and notch configurations. The filters are intended to minimize noise, motion artifact, and power line interference and therefore strain signal stability at the cost of amplitude. Second, older sensors such as the RS2334 are prone to limiting amplification to avoid clipping and saturation and therefore generate well-filtered but attenuated amplitude signals. Third, the integrity of the electrode-skin interface is of paramount importance. Dry skin, lack of conductive gel, or poorly attached electrodes can elevate contact impedance and therefore decrease signal amplitude, even if the system otherwise operates correctly.

In spite of these issues, the prototype consistently captured muscle activity and fatigue level changes. Real-time data recordings indicated that the electromyography (EMG) signal grew stronger on contraction and weaker on rest. Analog readings were then categorized into three zones: REST (<13), MODERATE FATIGUE (13 – 16), and HIGH FATIGUE (≥ 16). Each zone was assigned a specific color (GREEN, ORANGE, RED), and a buzzer warning was initiated during high-fatigue

intervals, thereby giving instant feedback to the user. This logic-based approach was useful in monitoring the build-up of fatigue during repetitive muscular activity.

The inherent operational efficiency of the system is in its ability to infer and interpret relative variations in EMG signal intensity rather than absolute voltage. The healthy operation of the buzzer at the right thresholds ensures the soundness of the underlying logic and the usefulness of the signal even with a constrained dynamic range. Moreover, this highlights the importance of monitoring relative signals in physiological monitoring, where there is subject variability and reduced signal amplitude in most cases.

From the performance viewpoint, the system possesses a number of benefits. It is economical, easy to construct, and lightweight—ideal for use in individual rehabilitation, athletic training, and ergonomic evaluation. Real-time monitoring of the levels of fatigue by serial output makes it easy for users or therapists to rapidly alter workload and recovery. From an engineering point of view, the system attains its basic objectives of real-time monitoring, ease of feedback, and functional precision with minimum hardware requirement.

There are some areas of future development, though. A dedicated post-EMG amplification stage can significantly improve the range and resolution of the analog signal. Digital filtering techniques such as Kalman filters or moving average can additionally smooth signal oscillations. Wireless communication modules (Wi-Fi or Bluetooth) can be incorporated for remote or mobile monitoring via smartphone apps. Additional sensors like IMUs or force sensors can provide multimodal feedback, enhancing system accuracy and context-awareness. In summary, the Muscle Fatigue Monitoring System performs its design functions well, offering a technically sound and practical method for the detection of fatigue. Even under low analog signals, the ability of the system to track signal trends and produce appropriate feedback serves to prove the low-voltage EMG outputs are not a limitation per se but a feature of well-filtered physiological signals. With more development, the prototype system can be shaped into a practical, wearable biofeedback system for use in the wider biomedical engineering, physiotherapy, and occupational safety communities.

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